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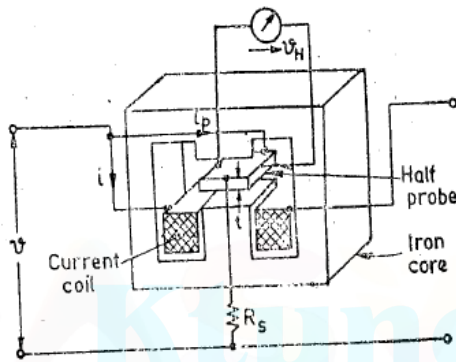
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Part A

Explain the working of Hall effect multipliers. [3 marks] (December 2021)

Hall Effect. If a strip of conducting material carries current in the presence of a transverse magnetic field as shown in Fig. , an emf is produced between two edges of conductor. The magnitude of the voltage depends upon the current, flux density and a property of conductor called "Hall Effect Co-efficient"



The current is passed through the current coil which produces a magnetic field proportional to the current i . This field is perpendicular to the Hall effect element. A current i_p , proportional to the voltage is passed through the Hall element in a direction perpendicular to field as shown. The current is limited by the multiplier resistance R_s . The output voltage of the Hall effect multiplier is :

$$v_H = R_H \frac{i_p B}{t}$$

where R_H = Hall co-efficient ; V-m/A-Wb m⁻²,
 B = flux density ; Wb/m², t = thickness of Hall element ; m.

$$\text{Now } B \propto i \text{ and } i_p = \frac{v}{R_s} \propto v.$$

$$\text{Hence } v_H \propto v i$$

∴ The output voltage of the Hall effect multiplier is proportional to instantaneous power.

2. What is a TOD meter? [3 marks] (December 2021)

The time Of Day (TOD) meter records the demand, time, and energy usage of electricity.

- TOD metering normally splits rates into an arrangement of multiple segments including on-peak, off-peak, and critical peak. The TOD metering benefits customers by providing reduced usage rates during off-peak times, which gives customers a chance to reduce their utility bills.
 - TOD metering involves dividing the day into tariff slots with higher rates at peak load periods and lower tariff rates at off-peak load periods. This can be used to automatically control usage on the part of the consumer.
 - It is often consumers' responsibility to control their own usage or pay accordingly.
-

The deflection produced by an Electrodynamometer wattmeter is proportional to the power being measured. Justify with necessary equations.

[3 marks] (December 2020) & [6 marks] (December 2020)

Instantaneous torque of the electro-dynamometer wattmeter is given by

$$T_i = i_p i_c \frac{dM}{d\theta}$$

Instantaneous value of voltage across the pressure-coil circuit is

$$v_p = \sqrt{2} \times V \sin \omega t$$

If the pressure coil resistance can be assumed to be very high, the whole pressure coil can be assumed to be behaving like a resistance only. The current i_p in the pressure coil thus, can be assumed to be in phase with the voltage v_p , and its instantaneous value is

$$i_p = \frac{v_p}{R_p} = \frac{\sqrt{2} \times V}{R_p} \sin \omega t = \sqrt{2} \times I_p \sin \omega t$$

where $I_p = V/R_p$ is the rms value of current in pressure coil.

If the current in the current coil lags the voltage in phase by an angle ϕ , instantaneous value of current through current coil is :

$$i_c = \sqrt{2} \times I \sin(\omega t - \phi)$$

Thus, instantaneous torque from can be written as

$$\begin{aligned} T_i &= \sqrt{2} \times I_p \sin \omega t \times \sqrt{2} \times I \sin(\omega t - \phi) \frac{dM}{d\theta} \\ &= 2I_p I \sin \omega t \sin(\omega t - \phi) \frac{dM}{d\theta} \\ &= I_p I \{ \cos \phi - \cos(2\omega t - \phi) \} \frac{dM}{d\theta} \end{aligned}$$

Presence of the term containing $2\omega t$, indicates the instantaneous torque varies at twice the frequency of voltage and current.

Average deflecting torque over a complete cycle is

$$\begin{aligned} T_d &= \frac{1}{T} \int_0^T T_i d\omega t = \frac{1}{2\pi} \int_0^{2\pi} I_p I [\cos \phi - \cos(2\omega t - \phi)] \frac{dM}{d\theta} d\omega t \\ &= \frac{I_p I}{2\pi} [\omega t \cos \phi]_0^{2\pi} \frac{dM}{d\theta} \\ &= I_p I \cos \phi \frac{dM}{d\theta} \\ &= \frac{V}{R_p} I \cos \phi \frac{dM}{d\theta} \\ &= \frac{VI \cos \phi}{R_p} \frac{dM}{d\theta} \end{aligned}$$

With a spring constant K , the controlling torque provided by the spring for a final steady-state deflection of θ is given by

$$T_C = K\theta$$

Under steady-state condition, the average deflecting torque will be balanced by the controlling torque provided by the spring. Thus, at balanced condition $T_C = T_d$

$$\begin{aligned} T_C &= T_d \\ K\theta &= \frac{VI \cos \phi}{R_p} \frac{dM}{d\theta} \\ \theta &= \frac{VI \cos \phi}{KR_p} \frac{dM}{d\theta} \\ \theta &= \left(K_1 \frac{dM}{d\theta} \right) P \end{aligned}$$

where, P is the power to be measured and $K_1 = 1/KR_p$ is a constant.

Steady-state deflection θ is thus found to be an indication of the power P to be measured.

In single phase induction type Energy meter, why shunt magnet flux should be in exact quadrature with the applied voltage? How this is made possible? [3 marks] (December 2020)

Energy meter will register true energy only if angle between flux ϕ and supply voltage V is exactly 90 degree. But because the pressure coil is not entirely inductive and may have some resistance this angle may be slightly less than 90 degree.

Suitable adjustments can be implemented such that the shunt magnet flux linking with the disc can be made to lag the supply voltage by an angle exactly equal to 90°.

Shading Coil with Adjustable Resistance

Figure (a) below shows the arrangement where an additional coil (shading coil) with adjustable resistance is placed on the central limb of the shunt magnet close to the disc.

Main flux created by the shunt magnet induces an emf in this shading coil.

This emf creates its own flux. These two fluxes result in a modified flux to pass through the air gap to link the disc and thus produce the driving torque. With proper adjustment of the shading coil resistance, the resultant flux can be made to lag the supply voltage exactly by an angle of 90° .

A similar result of lag adjustment can be obtained by the use of copper shading bands placed on the central limb of the shunt magnet. Fig (b)

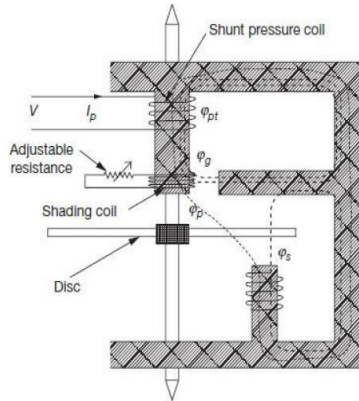


Fig (a)

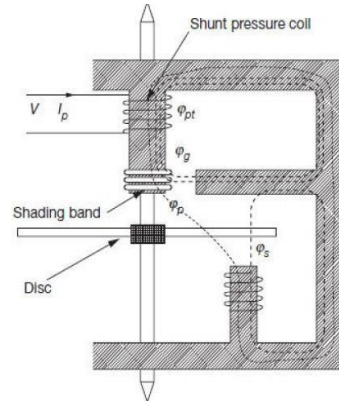
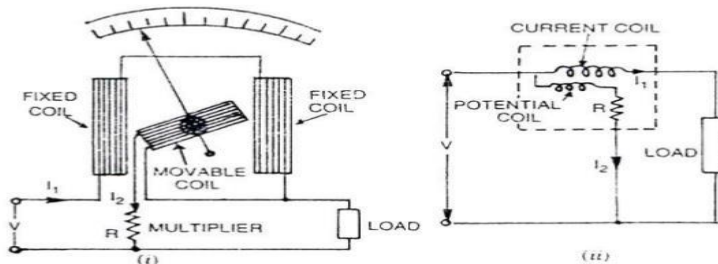


Fig (b)

PART B

With neat diagram explain the construction and working of an electrodynamic wattmeter.
[8 marks] (December 2021)

Single phase electro-dynamometer (electro-dynamic) type wattmeter



Fixed coils

Such an instrument has two coils connected in different ways to the same circuit of which power is to be measured.

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Module II: University Questions and Answers

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The *fixed coils* or the *field coils* are connected in series with the load so as to carry the same current as the load. The fixed coils are hence, termed as the *Current Coils (CC)* of the wattmeter.

The main magnetic field is produced by these fixed coils.

This coil is divided into two sections so as to provide a more uniform magnetic field near the centre and to allow placement of the instrument's moving shaft.

Fixed coils are usually wound with thick wires for carrying the main load current through them. Windings of the fixed coil is normally made of stranded conductors running together but, insulated from each other.

Moving Coil System

The **moving coil** that is connected across the load carries a current proportional to the voltage.

Since the moving coil carries a current proportional to the voltage, it is called the *voltage coil* or the *pressure coil*, or simply *PC* of the wattmeter.

The moving coil is entirely embraced by the pair of fixed coils.

A high-value **non-inductive resistance** is connected in series with the voltage coil to restrict the current through it to a small value, and also to ensure that the voltage coil current remains as far as possible in phase with the load voltage.

The moving coil, made of fine wires, is wound either as a self-sustaining air-cored coil or else wound on a non-metallic former.

Control. Spring control is used for the instrument.

Damping. Air friction damping is used. The moving system carries a light aluminium vane which moves in a sector shaped box.

Scales and Pointers. They are equipped with mirror type scales and knife edge pointers to remove reading errors due to parallax.

Principle of Operation:

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V = voltage to be measured (rms)

I = current to be measured (rms)

i_P = voltage (pressure) coil instantaneous current

i_C = current coil instantaneous current

R_V = external resistance connected with pressure coil

R_P = resistance of pressure coil circuit (PC resistance + R_V)

M = mutual inductance between current coil and pressure coil

θ = angle of deflection of the moving system

ω = angular frequency of supply in radians per second

ϕ = phase-angle lag of current I with respect to voltage V

The instantaneous torque of the electro-dynamometer wattmeter is given by

$$T_I = i_P i_C \frac{dM}{d\theta}$$

Instantaneous value of voltage across the pressure-coil circuit is

$$v_P = \sqrt{2} \times V \sin \omega t$$

If the pressure coil resistance can be assumed to be very high, the whole pressure coil can be assumed to be behaving like a resistance only. The current i_P in the pressure coil thus, can be assumed to be in phase with the voltage v_P , and its instantaneous value is

$$i_P = \frac{v_P}{R_P} = \frac{\sqrt{2} \times V}{R_P} \sin \omega t = \sqrt{2} \times I_P \sin \omega t$$

where $I_P = V/R_P$ is the rms value of current in pressure coil.

If the current in the current coil lags the voltage in phase by an angle ϕ , instantaneous value of current through current coil is :

If the current in the current coil lags the voltage in phase by an angle ϕ , instantaneous value of current through current coil is :

$$i_c = \sqrt{2} \times I \sin(\omega t - \phi)$$

Thus, instantaneous torque from (1) can be written as

$$\begin{aligned} T_i &= \sqrt{2} \times I_p \sin \omega t \times \sqrt{2} \times I \sin(\omega t - \phi) \frac{dM}{d\theta} \\ &= 2I_p I \sin \omega t \sin(\omega t - \phi) \frac{dM}{d\theta} \\ &= I_p I \{ \cos \phi - \cos(2\omega t - \phi) \} \frac{dM}{d\theta} \end{aligned}$$

Presence of the term containing $2\omega t$, indicates the instantaneous torque varies at twice the frequency of voltage and current.

Average deflecting torque over a complete cycle is

$$\begin{aligned} T_d &= \frac{1}{T} \int_0^T T_i d\omega t = \frac{1}{2\pi} \int_0^{2\pi} I_p I \{ \cos \phi - \cos(2\omega t - \phi) \} \frac{dM}{d\theta} d\omega t \\ &= \frac{I_p I}{2\pi} [\omega t \cos \phi]_0^{2\pi} \frac{dM}{d\theta} \\ &= I_p I \cos \phi \frac{dM}{d\theta} \\ &= \frac{V}{R_p} I \cos \phi \frac{dM}{d\theta} \\ &= \frac{VI \cos \phi}{R_p} \frac{dM}{d\theta} \end{aligned}$$

With a spring constant K , the controlling torque provided by the spring for a final steady-state deflection of θ is given by

$$T_C = K\theta$$

Under steady-state condition, the average deflecting torque will be balanced by the controlling torque provided by the spring. Thus, at balanced condition $T_C = T_d$

$$\begin{aligned}T_C &= T_d \\K\theta &= \frac{VI \cos \phi}{R_p} \frac{dM}{d\theta} \\ \theta &= \frac{VI \cos \phi}{KR_p} \frac{dM}{d\theta} \\ \theta &= \left(K_1 \frac{dM}{d\theta} \right) P\end{aligned}$$

where, P is the power to be measured and $K_1 = 1/KR_p$ is a constant.

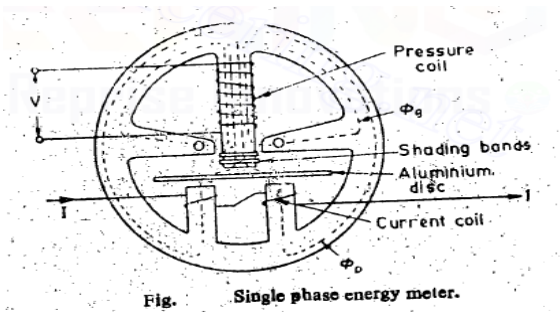
Steady-state deflection θ is thus found to be an indication of the power P to be measured.

Derive the expression for the torque of a single phase induction type energy meter

[6 marks] (December 2021)

In all induction-type instruments, two time-varying fluxes are created in the windings provided on the static part of the instrument. These fluxes are made to link with a metal disc or drum and produce emf therein. These emfs in turn, circulate eddy current on the body of the metal disc.

Interaction of these fluxes and eddy currents produce torques that make the disc or drum to rotate. closed-sector-shaped chamber.



Let V = applied voltage,

I = load current,

ϕ = phase angle of load,

I_p = pressure coil current,

Δ = phase angle between supply voltage and pressure coil flux. f = frequency,

Z = impedance of eddy current paths,

α = phase angle of eddy current paths,

E_{ep} = eddy emf induced by flux Φ_p ,

I_{ep} = eddy current due to flux Φ_p ,

E_{es} = eddy emf induced by flux Φ_s ,

I_{es} = eddy current due to flux Φ_s .

From Eqn. 8'97, net driving torque

$$T_d \propto \Phi_1 \Phi_2 \frac{f}{Z} \sin \beta \cos \alpha = K_1 \Phi_1 \Phi_2 \frac{f}{Z} \sin \beta \cos \alpha$$

where

K_1 = a constant.

Now β = phase angle between fluxes Φ_1 and Φ_2 .

In our case, the two fluxes are Φ_p and Φ_s ,

$\therefore \beta$ = phase angle between fluxes Φ_p and Φ_s = $(\Delta - \phi)$

$$\therefore \text{Driving torque } T_d = K_1 \Phi_p \Phi_s \frac{f}{Z} \sin (\Delta - \phi) \cos \alpha$$

But $\Phi_p \propto V$ and $\Phi_s \propto I$.

$$\therefore T_d = K_2 VI \frac{f}{Z} \sin (\Delta - \phi) \cos \alpha$$

If f , Z and α are constant, $T_d = K_3 VI \sin (\Delta - \phi)$

If N is the steady speed, braking torque $T_b = K_4 N$.

At steady speed the driving torque must equal the braking torque

$$\therefore K_4 N = K_3 VI \sin (\Delta - \phi), N = K VI \sin (\Delta - \phi)$$

$$\text{If } \Delta = 90^\circ, N = K VI \sin (90 - \phi) = K VI \cos \phi \\ = K \times (\text{power}).$$

Thus in order that the speed of rotation is proportional to power, angle Δ should be equal to 90° . Hence the flux Φ_p must be made to lag the supply voltage by exactly 90° .

$$\text{Total number of revolutions} = \int N dt = K \int VI \sin (\Delta - \phi) dt$$

$$\text{If } \Delta = 90^\circ, \text{ total number of revolutions} = K \int (VI \cos \phi) dt$$

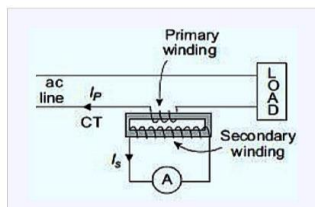
$$= K \int (\text{power}) dt = K \times (\text{energy})$$

Derive the expression for ratio and phase angle error in a current transformer

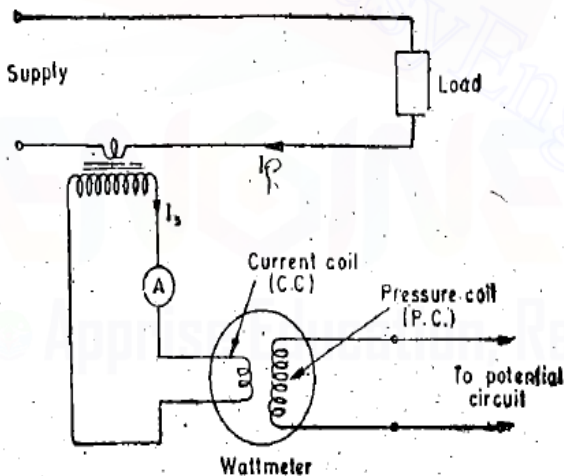
[10 marks] (December 2021) & [14 marks] (December 2020)

The primary winding of a current transformer is connected in series with the line carrying the main current. The secondary winding of the CT, where the current is many times stepped down, is directly connected across an ammeter, for measurement of current; or across the current coil of a wattmeter, for measurement of power; or across the current coil of a watt-hour meter for measurement of energy; or across a relay coil.

Current Transformer for the measurement of current



$$n = \text{turns ratio} = \frac{\text{number of secondary winding turns}}{\text{number of primary winding turns}}$$



r_s = resistance of the secondary winding,
 x_s = reactance of the secondary winding,
 r_b = resistance of external burden i.e., resistance of meters current coils etc. including leads,
 x_b = reactance of external burden i.e., reactance of meters, current coils, etc. including leads,
 E_p = primary induced voltage,
 E_s = secondary induced voltage,

Use of C.T. for current and power measurement .

N_p = number of primary winding turns, N_s = number of secondary winding turns,
 V_s = voltage at the secondary winding terminals, I_s = secondary winding current,
 I_p = primary winding current, θ = phase angle of transformer,
 Φ = working flux of the transformer,
 δ = angle between secondary induced voltage and secondary current,

= phase angle of total burden including impedance of secondary winding $= \tan^{-1} \left(\frac{x_s + x_e}{r_s + r_e} \right)$

Δ = phase angle of secondary load circuit i.e., of external burden $= \tan^{-1} \frac{x_s}{r_s}$,

I_0 = exciting current,

I_m = magnetising component of exciting current, I_e = loss component of exciting current,

Transformation Ratio. Consider a small section of the phasor diagram as shown in Fig.

We have $\angle bac = 90^\circ - \delta - \alpha$, $ac = I_0$, $oa = nI_s$ and $oc = I_p$

$\therefore bc = I_0 \sin(90^\circ - \delta - \alpha) = I_0 \cos(\delta + \alpha)$, $ab = I_0 \cos(90^\circ - \delta - \alpha) = I_0 \sin(\delta + \alpha)$.

Now $(oc)^2 = (oa + ab)^2 + (bc)^2$ or $I_p^2 = [nI_s + I_0 \sin(\delta + \alpha)]^2 + [I_0 \cos(\delta + \alpha)]^2$

$$= n^2 I_s^2 + I_0^2 \sin^2(\delta + \alpha) + 2nI_s I_0 \sin(\delta + \alpha) + I_0^2 \cos^2(\delta + \alpha)$$

$$= n^2 I_s^2 + 2nI_s I_0 \sin(\delta + \alpha) + I_0^2$$

$$\therefore I_p = [n^2 I_s^2 + 2nI_s I_0 \sin(\delta + \alpha) + I_0^2]^{1/2}$$

Transformation ratio

$$R = \frac{I_p}{I_s} = \frac{[n^2 I_s^2 + 2nI_s I_0 \sin(\delta + \alpha) + I_0^2]^{1/2}}{I_s}$$

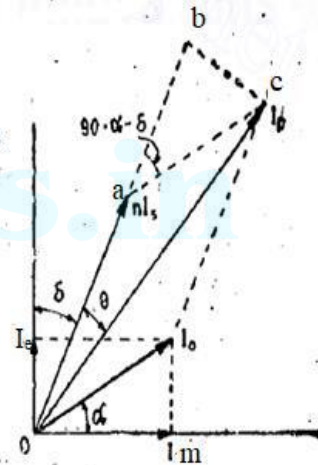
Now in a well designed current transformer $I_0 \ll nI_s$.
Usually I_0 is less than 1 per cent of I_p and I_p is, therefore, very nearly equal to nI_s .

$\therefore$ Eqn. can be written as

$$R \approx \frac{[n^2 I_s^2 + 2nI_s I_0 \sin(\delta + \alpha) + I_0^2 \sin^2(\delta + \alpha)]^{1/2}}{I_s}$$

$$\approx \frac{nI_s + I_0 \sin(\delta + \alpha)}{I_s} \approx n + \frac{I_0}{I_s} \sin(\delta + \alpha)$$

Although only approximate, Eqn. is sufficiently accurate for practically all purposes. The above theory is applicable to case when the secondary burden has a lagging power factor i.e., when the burden is inductive which is normally the case.



.. A section of phasor diagram of a C.T.

Eqn. can be further expanded as :

$$R \approx n + \frac{I_0}{I_s} (\sin \delta \cos \alpha + \cos \delta \sin \alpha) \approx n + \frac{I_m \sin \delta + I_e \cos \delta}{I_s}$$

$$I_m = I_0 \cos \alpha \text{ and } I_e = I_0 \sin \alpha$$

Phase Angle. The angle by which the secondary current phasor, when reversed, differs in phase from the primary current, is known as the phase angle of the transformer.

This angle is taken to be +ve if the secondary current reversed leads the primary current. The angle is taken as -ve if secondary current reversed lags behind the primary current.

The angle between I_s reversed and I_p is θ . Therefore, the phase angle is θ .

From the phasor diagram $\tan \theta = \frac{bc}{ob} = \frac{bc}{oa+ab} = \frac{I_0 \cos(\delta+\alpha)}{nI_s + I_0 \sin(\delta+\alpha)}$

As θ is very small, we can write $\theta = \frac{I_0 \cos(\delta+\alpha)}{nI_s + I_0 \sin(\delta+\alpha)} \text{ rad.}$

Now I_0 is very small as compared to nI_s and therefore, we can neglect the term $I_0 \sin(\delta+\alpha)$

$\therefore \theta \approx \frac{I_0 \cos(\delta+\alpha)}{nI_s} \text{ rad.}$

$\approx \frac{I_0 \cos \delta \cos \alpha - I_0 \sin \delta \sin \alpha}{nI_s} \approx \frac{I_m \cos \delta - I_e \sin \delta}{nI_s} \text{ rad.}$

$\approx \frac{180}{\pi} \left(\frac{I_m \cos \delta - I_e \sin \delta}{nI_s} \right) \text{ degree}$

Ratio error. Ratio Error is defined as :

Percentage ratio error = $\frac{\text{nominal ratio} - \text{actual ratio}}{\text{actual ratio}} \times 100$
 $= \frac{K_n - R}{R} \times 100$

Phase angle error. Phase angle $\theta = \frac{180}{\pi} \left[\frac{I_m \cos \delta - I_e \sin \delta}{nI_s} \right] \text{ degree}$

Approximate formulae for errors. The usual instrument burden is largely resistive with some inductance and therefore, δ is positive and is generally small.

Hence $\sin \delta \approx 0$ and $\cos \delta \approx 1$. Therefore, we can write

$R \approx n + \frac{I_e}{I_s}$

and

$\theta \approx \frac{180}{\pi} \left[\frac{I_m}{nI_s} \right] \text{ degree}$

But $I_p \approx nI_s$ and therefore,

$R \approx n + \frac{nI_e}{I_p} = n \left(1 + \frac{I_e}{I_p} \right)$

and

$\theta = \frac{180}{\pi} \frac{I_m}{I_p} \text{ degree}$

Explain how power can be measured in a 3-phase circuit using two wattmeters with a neat circuit diagram in unbalanced load conditions.

[4 marks] (December 2021)

The connections for measurement of power in the case of a star-connected three-phase load are shown

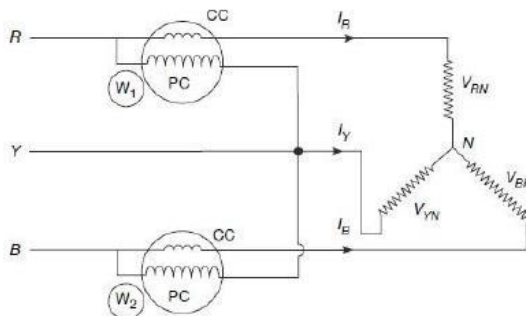
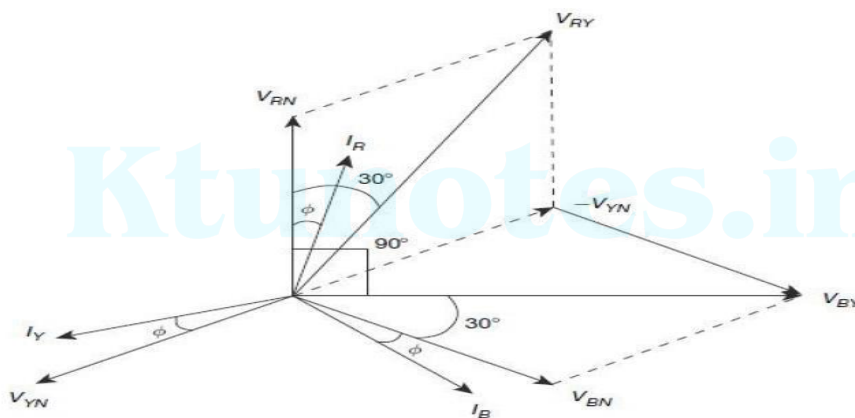


Figure Two-wattmeter method for star-connected load



Let, V_{RN} , V_{BN} , and V_{YN} are phase voltages and I_R , I_B , and I_Y are phase currents for the balanced three phase star-connected system under study.

For a balanced system, phase voltages, $V_{RN} = V_{BN} = V_{YN} = V$ (say)

And, phase currents, $I_R = I_B = I_Y = I$ (say)

For a star-connected system,

Line voltages $V_{RY} = V_{YB} = V_{BR} = \sqrt{3} V$

Line currents $I_R = I_B = I_Y = I$

Power factor = $\cos \phi$, where ϕ is the angle by which each of the phase currents lag the corresponding phase voltages.

Current through the CC of wattmeter W_1 is I_R and voltage across its potential coil is V_{RY} . The current I_R leads the voltage by V_{RY} an angle $(30^\circ - \phi)$, as shown in Figure

$\therefore$ reading of wattmeter W_1 is,

$$P_1 = V_{RY} \times I_R \cos(30^\circ - \phi) = \sqrt{3} VI \cos(30^\circ - \phi)$$

Current through the CC of wattmeter W_2 is I_B and voltage across its potential coil is V_{BY} . The current I_B lags the voltage by V_{BY} an angle $(30^\circ + \phi)$, as shown in Figure

$\therefore$ reading of wattmeter W_2 is,

$$P_2 = V_{BY} \times I_B \cos(30^\circ + \phi) = \sqrt{3} VI \cos(30^\circ + \phi)$$

Sum of these two-wattmeter readings:

$$P_1 + P_2 = \sqrt{3} VI \cos(30^\circ - \phi) + \sqrt{3} VI \cos(30^\circ + \phi) = 3 VI \cos \phi$$

This is the total power consumed by the load, adding together the three individual phases. Thus, at any power factor, the total power consumed by the load will be, in any case, summation of the two wattmeter readings.

Explain the various errors in electro-dynamometer type wattmeter. [8 marks] (December 2020)

Error due to Pressure-Coil Inductance

Error due to Pressure Coil Capacitance

Error due to Connection

Eddy-current Errors

Stray Magnetic Field Errors

Error Caused by Vibration of the Moving System

Temperature Errors

1. Error due to Pressure-Coil Inductance

It was assumed during the discussions so far that the pressure coil circuit is purely resistive. In reality, however, the pressure coil will have certain inductance along with resistance. This will introduce errors in measurement unless necessary compensations are taken care of

V = voltage applied to the pressure coil circuit (rms) , I = current in the current coil circuit (rms)

I_P = current in the voltage (pressure) coil circuit (rms), r_P = resistance of pressure coil only

L = inductance of pressure coil, R_V = external resistance connected with pressure coil

R_P = resistance of pressure coil circuit (PC resistance + R_V), Z_P = impedance of pressure coil circuit

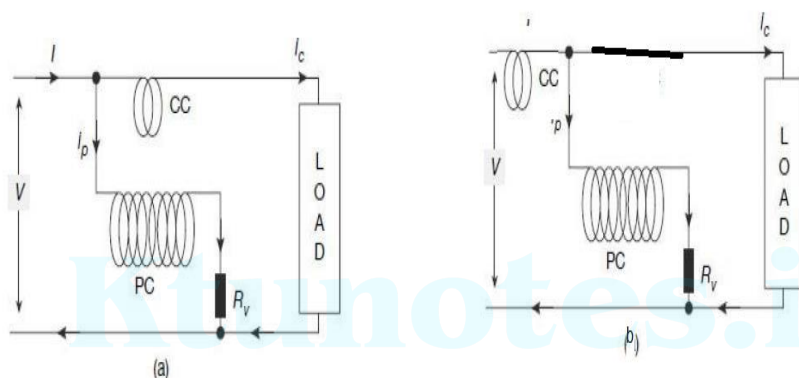
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The effect of stray capacitance of the pressure coil is opposite to that due to inductance. Therefore, the wattmeter reads low on lagging power factors and high on leading power factors of the load. Actual reading of the wattmeter, thus, once again needs to be corrected by the corresponding correction factors to obtain the true reading.

The effect of capacitance (as well as inductance) varies with variable frequency of the supply.

3. Error due to Connection



There are two alternate methods of connection of wattmeter to the circuit for measurement of power. These are shown in Figure. In either of these connection modes, errors are introduced in measurement due power losses in pressure coil and current coil.

In the connection of (a), the pressure coil is connected across the supply, thus pressure coil measures the voltage across the load, plus the voltage drop across the current coil. Wattmeter reading in this case will thus include power loss in current coil as well, along with power consumed by the load.

Wattmeter reading = Power consumed by load + Power loss in CC

In the connection of (b), the current coil is connected to the supply side; therefore, it carries load current plus the pressure coil current. Hence, wattmeter reading in this case includes, along with power consumed by the load, power loss in the pressure coil as well.

Wattmeter reading = Power consumed by load + Power loss in PC

In the case when load current is small, power loss in the current coil is small and hence the connection of Figure (a) will introduce comparatively less error in measurement.

On the other hand, when load current is large, current branching through the pressure coil is relatively small and error in measurement will be less if connection of Figure (b) is used.

Eddy-current Errors

Unless adequate precautions are adopted, eddy-currents may be induced in metallic parts of the instrument and even within the thickness of the conductors by the alternating magnetic field of the current coil.

These Eddy-currents produce spurious magnetic fields of their own and distort the magnitude and phase of the main current coil magnetic field, thereby introducing errors in the measurement of power.

Error caused by eddy currents is not easy to estimate and may become objectionable if metal parts are not carefully avoided from near the current coil.

In fact, solid metal in coil supports and structural parts should be kept to a minimum as far as practicable.

Any metal that is used is kept away and is selected to have high resistivity so as to reduce eddy currents induced in it.

5. Stray Magnetic Field Errors

The operating field in electro-dynamometer-type instruments is weak, special care must be taken to protect these instruments from external magnetic fields. Hence, these instruments should be shielded against the effects of stray magnetic fields.

Laminated iron shields are used in portable laboratory instruments, while steel casings are provided as shields in switchboard-mounted wattmeter.

6. Error Caused by Vibration of the Moving System

The instantaneous torque on the moving system varies cyclically at twice the frequency of the

$$\text{voltage and current. } T_i = I_p I \{ \cos \phi - \cos(2\omega t - \phi) \} \frac{dM}{d\theta}$$

If any part of the moving system, such as the spring or the pointer has a natural frequency close to that of torque pulsation, then accidental resonance may take place. In such a case, the moving system may vibrate with considerable amplitude.

These vibrations may pose problems while noting the pointer position on the scale.

These errors due to vibrations may be avoided by designing the moving elements to have natural frequencies much further away from twice the frequency of the supply voltage.

7. Temperature Errors

Temperature changes may affect the accuracy of the wattmeter by altering the coil resistances.

The temperature may change due to changes in room temperature or even due to heating effects in conductors with the flow of current. Change in temperature also affects the spring stiffness, thereby introducing errors in the deflection process.

High-precision instruments are fitted with temperature-compensating resistors that tend to neutralize the effects of temperature variation.